

# Data Cleaning & Preparation — Change Log

DecodeLabs Industrial Training Kit · Project 1 · Batch 2026

This log documents the audit performed on the raw orders dataset (1,200 records, 14 fields) and every corrective action taken before it was certified as analysis-ready. Per the verification gate for Project 1, the dataset must reach a 0% error rate on unique identifiers and date formats before Project 2 can begin.

## Phase 1 — Strategic Imputation (Missing Values)

Every column was checked for nulls. Only **CouponCode** contained blanks (309 of 1,200 rows, ~25.75%). This was diagnosed as a legitimate business state — a coupon was simply not applied at checkout — rather than a data-entry error, since no other field on those rows was affected. Deleting these rows would have discarded valid revenue data (listwise deletion), so the blanks were explicitly imputed as the label **'No Coupon'** instead of a statistical measure like mean/median/mode, which does not apply to a categorical code.

## Phase 2 — The Integrity Audit (Duplicates)

The dataset was checked for full-row duplicates and for duplicate values in the two fields that must be unique per business rule: **OrderID** and **TrackingNumber**. All three checks returned zero duplicates, so no rows were removed.

## Phase 3 — Speak One Language (Format Standardization)

Dates were normalized to ISO 8601 (YYYY-MM-DD); all 1,200 values parsed cleanly with zero invalid entries. All text fields were checked for leading/trailing whitespace and casing inconsistencies (e.g. city/category name variants) — none were found. Currency fields (UnitPrice, TotalPrice) were standardized to 2 decimal places, and every TotalPrice was cross-checked against Quantity x UnitPrice — all 1,200 rows reconciled exactly.

## Change Log

| Change ID | Description                                             | Impact                  | Status   |
|-----------|---------------------------------------------------------|-------------------------|----------|
| CR001     | Imputed 'CouponCode' blanks as 'No Coupon'              | 309 records clarified   | Resolved |
| CR002     | Standardized 'Date' to ISO 8601 (YYYY-MM-DD)            | 1,200 records verified  | Resolved |
| CR003     | Verified uniqueness of OrderID & TrackingNumber         | 0 duplicates found      | Verified |
| CR004     | Cross-checked TotalPrice = Quantity x UnitPrice         | 0 mismatches found      | Verified |
| CR005     | Standardized currency fields to 2 decimal places        | 1,200 records formatted | Resolved |
| CR006     | Audited categorical fields for spelling/casing variants | 0 inconsistencies found | Verified |

## Verification Gate — Project 1 Threshold

| Metric                                                      | Result |
|-------------------------------------------------------------|--------|
| Error rate on unique identifiers (OrderID / TrackingNumber) | 0%     |

|                            |      |
|----------------------------|------|
| Error rate on date formats | 0%   |
| Rows before cleaning       | 1200 |
| Rows after cleaning        | 1200 |

Result: The dataset meets the 0% error threshold on unique identifiers and date formats required to unlock Project 2.

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Prepared as part of DecodeLabs Data Analytics Industrial Training Kit — Project 1. Deliverables: Cleaned\_Dataset\_Project1.xlsx (cleaned data + audit sheet), Change\_Log\_Project1.pdf.